KATHMANDU MODEL COLLEGE

Department of BBA, BBM & BCA

ARTIFICIAL INTELLIGENCE & PROMPT ENGINEERING

COMPLETE BEGINNER'S GUIDE

From Zero to AI-Powered Professional

Workshop Details	
Program	BBM (8–9 Chaitra) | BBA (10–11 Chaitra) | BCA (15–16 Chaitra)
Duration	2 Days × 4 Hours = 8 Hours Total
Tools Covered	ChatGPT, Gemini, Claude, Gamma.app, Leonardo.ai, Perplexity
Level	Complete Beginner → Intermediate
Edition	2025 AI Workshop Series

Chaitra 2081 | Kathmandu, Nepal

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CHAPTER 1: Introduction to Artificial Intelligence

"Artificial Intelligence is the new electricity. Just as electricity transformed industries 100 years ago, AI will transform virtually every major industry in the coming decades." — Andrew Ng, AI Pioneer

1.1 What Is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence processes by computer systems. These processes include learning (acquiring information and rules for using it), reasoning (using rules to reach approximate or definite conclusions), and self-correction. AI systems can perform tasks that traditionally required human intelligence such as visual perception, speech recognition, decision-making, and natural language understanding.

In simple terms, AI is a computer program that can "think" and "learn" to solve problems. Instead of following strict rules written by a programmer for every situation, an AI learns patterns from large amounts of data and uses those patterns to make decisions on new, unseen situations.

1.2 A Brief History of AI

Year	Milestone	Significance
1950	Alan Turing proposes the Turing Test	First formal concept of machine intelligence
1956	Dartmouth Conference – AI coined as a term	Birth of AI as an academic discipline
1966	ELIZA chatbot created at MIT	First natural language processing program
1997	IBM Deep Blue defeats chess champion Kasparov	AI beats humans at complex strategy games
2011	IBM Watson wins Jeopardy!	AI understands natural language at human level
2012	Deep Learning revolution (ImageNet)	Neural networks achieve human-level image recognition
2016	AlphaGo defeats world Go champion	AI masters one of the most complex games
2017	Transformer architecture introduced (Google)	Foundation of all modern LLMs
2020	GPT-3 released by OpenAI	AI can write human-like text at scale
2022	ChatGPT launches – 1M users in 5 days	AI becomes mainstream consumer technology
2023	GPT-4, Gemini, Claude 2 released	Multimodal AI understands images + text

2024–25	AI agents, multimodal reasoning, local AI	AI integrated into every workflow

1.3 Types of AI: Understanding the Landscape

1.3.1 By Capability

AI can be classified based on its capabilities into three broad categories:

- Narrow AI (Weak AI): Designed for a specific task. Examples include Siri, Google Translate, spam filters, and recommendation systems on Netflix or YouTube. All AI systems we use today fall into this category. They are extremely good at one thing but cannot generalize.
- General AI (AGI – Artificial General Intelligence): A theoretical AI that can perform any intellectual task that a human can. It doesn't exist yet but is the long-term goal of many AI researchers. AGI would be able to reason, learn, and adapt across any domain.
- Super AI (ASI – Artificial Super Intelligence): A hypothetical AI that surpasses human intelligence in all aspects. This remains firmly in the realm of science fiction and philosophical debate. It is the subject of significant ethical concern.

1.3.2 By Approach

AI Type	How It Works	Examples	Strength
Rule-Based AI	Follows pre-written if-then rules	Chess engines (early), ATM systems	Predictable, transparent
Machine Learning	Learns patterns from data	Email spam filter, fraud detection	Adapts to new data
Deep Learning	Neural networks with many layers	Image recognition, voice assistants	Handles complex patterns
Generative AI	Creates new content from prompts	ChatGPT, Midjourney, Suno AI	Creative, versatile
Reinforcement Learning	Learns by trial & error with rewards	AlphaGo, robotics, game AI	Masters complex strategies

1.4 Generative AI vs. Traditional AI

This is the most important distinction for this workshop. Traditional AI systems are designed to analyze and classify — they take inputs and produce predictions or decisions. Generative AI systems are designed to create — they take prompts (instructions) and produce new content such as text, images, code, music, or video.

❌ Traditional AI (Analytical)	✅ Generative AI (Creative)
• Analyzes existing data	• Creates new, original content
• Classifies, predicts, or recommends	• Generates text, images, code, audio
• Output: a label, number, or decision	• Output: a document, image, or solution

• Example: 'This email is spam (95% confidence)' • Example: 'This tumor is malignant' • Example: 'Customer likely to churn' • Requires labeled training data • Deterministic and reproducible	• Example: Write a marketing email for our product • Example: Explain diabetes in simple terms • Example: Build a website for our startup • Learns from vast unlabeled data • Probabilistic and creative

1.5 How Does AI "Learn"? Machine Learning Explained

Machine learning is the engine behind most modern AI. Instead of being explicitly programmed for every scenario, a machine learning model learns from examples. Imagine teaching a child to recognize cats — you show them hundreds of pictures of cats and non-cats. Eventually, they learn the features: ears, whiskers, tail, fur. Machine learning works the same way.

The Three Types of Machine Learning

1. Supervised Learning: The model learns from labeled data (input + correct answer pairs). Like a student with an answer key. Used for: email spam detection, medical diagnosis, price prediction.
2. Unsupervised Learning: The model finds patterns in unlabeled data without being told what to look for. Like grouping similar objects without knowing their names. Used for: customer segmentation, anomaly detection, data compression.
3. Reinforcement Learning: The model learns by taking actions in an environment and receiving rewards or penalties. Like training a dog with treats. Used for: game-playing AI, robot navigation, recommendation optimization.

1.6 Neural Networks & Deep Learning

Deep Learning is a subset of Machine Learning that uses artificial neural networks — computational systems inspired by the human brain. A neural network consists of layers of nodes (neurons) connected by weighted edges. Information flows through these layers and is transformed at each step.

Understanding Neural Network Layers

Layer Type	Function	Analogy
Input Layer	Receives raw data (text, pixels, audio)	Your eyes/ears receiving information
Hidden Layers	Transform and extract features progressively	Your brain processing and interpreting signals
Output Layer	Produces the final prediction or generation	Your mouth/hands producing a response

Deep Learning excels when there is a massive amount of training data and sufficient computing power. It is the technology behind image recognition, speech synthesis, translation, and all modern Large Language Models.

1.7 The AI Ecosystem: Tools You'll Use in This Workshop

Tool	Category	Primary Use	Access
ChatGPT (OpenAI)	Generative AI / LLM	Text generation, coding, analysis, brainstorming	chat.openai.com
Google Gemini	Generative AI / LLM	Research, multimodal tasks, Google integration	gemini.google.com
Claude (Anthropic)	Generative AI / LLM	Long-form writing, reasoning, coding, safety	claude.ai
Perplexity AI	AI Search Engine	Research, citations, real-time web search	perplexity.ai
Gamma.app	AI Presentation Tool	Creating professional slide decks from text	gamma.app
Leonardo.ai	AI Image Generator	Logos, visuals, marketing assets, branding	leonardo.ai
GitHub Copilot	AI Coding Assistant	Code completion, debugging, code explanation	github.com/copilot
Cursor	AI Code Editor	Full IDE with AI pair programmer	cursor.sh

1.8 Real-World AI Applications in Nepal & South Asia

AI is not just a Silicon Valley phenomenon. It is transforming industries right here in Nepal and across South Asia. Understanding these local applications makes AI relevant and actionable for you as students.

Business & Commerce

- E-commerce platforms like Daraz use AI for product recommendations and price optimization
- Banks like NIC Asia and Global IME use AI for fraud detection and credit scoring
- Ride-sharing platforms use AI for dynamic pricing, route optimization, and demand forecasting
- Nepal Airlines and tourism operators use AI chatbots for booking assistance

Agriculture & Environment

- AI-powered drones are being tested for crop monitoring in the Terai region
- Flood prediction models using AI help communities in flood-prone areas prepare
- Climate change research institutions use ML to analyze Himalayan glacier data

Healthcare

- Telemedicine platforms use AI to pre-screen patient symptoms
- Radiology AI helps rural health posts analyze X-rays without specialist doctors

- Mental health apps use AI-driven conversations for initial counseling support

Education

- Personalized learning platforms adapt content to individual student performance
- AI translation tools help bridge language barriers across Nepal's linguistic diversity
- Automated grading systems save teachers time on routine assessments

1.9 Understanding AI Limitations & Hallucinations

Before you start relying on AI tools, you MUST understand their fundamental limitations. The most dangerous mistake a beginner can make is trusting AI output without verification.

What Are AI Hallucinations?

An AI hallucination occurs when an AI system confidently produces information that is factually incorrect, fabricated, or completely made up. The AI is not lying — it genuinely cannot tell truth from fiction. It is pattern-matching and generating statistically plausible text, not retrieving verified facts from a database.

⚠️ REAL HALLUCINATION EXAMPLES — Always Verify These Types of Claims
❌ 'According to a 2023 study by Harvard Business School published in the Journal of Applied Economics...'
→ This citation may be entirely fabricated. The journal, authors, and findings may not exist.
❌ 'The Nepal Tourism Board reported 1.2 million visitors in Q3 2024...'
→ AI may invent specific statistics. Always verify with official NTB reports.
❌ 'Python's pandas library introduced the .smart_fill() method in version 2.1...'
→ This method does not exist. AI often invents plausible-sounding but fake function names.
❌ 'Mount Everest was first climbed in 1948 by Edmund Hillary...'

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   → The correct date is 1953. AI can get well-known facts wrong under
certain conditions.
```

Why Do Hallucinations Happen?

4. LLMs predict the next most likely token (word piece) based on patterns — they do not look up facts in a database
5. Training data has errors, biases, and outdated information that the model absorbed
6. The model was never designed to say 'I don't know' — it tries to produce an answer
7. Rare topics have sparse training data, so the model 'fills in' with plausible guesses
8. The model cannot distinguish between correlation and causation in its training data

How to Detect and Handle Hallucinations

✅ THE VERIFY-BEFORE-USE PROTOCOL
1. For any specific statistic cited by AI → search official sources (government sites, academic journals)
2. For any research paper or book → search on Google Scholar or Research Gate to confirm it exists
3. For any date, name, or event → cross-check with Wikipedia and at least one primary source
4. For any code snippet → run and test it in your actual environment before using in production
5. Use Perplexity AI (covered in Chapter 9) which cites sources automatically for fact-checking
6. When in doubt, ask the AI: 'How confident are you about this? What sources support this claim?'

1.10 Academic Integrity & Ethical AI Use

Using AI tools does not automatically constitute cheating or plagiarism — but using AI irresponsibly can. As students of KMC, you must understand the ethical boundaries clearly. The rule is simple: AI is a tool to enhance your thinking, not replace it.

Scenario	Ethical?	Why
Using AI to brainstorm ideas for your thesis topic	✅ Yes	AI helps you think, you develop the ideas

Asking AI to explain a concept you don't understand	✅ Yes	Learning from AI like using a textbook
Using AI to check grammar in your essay	✅ Yes	Tool-assisted editing, you wrote the content
Asking AI to find and summarize research sources (then you verify)	✅ Yes	Research assistance, you verify and synthesize
Submitting AI-generated text verbatim as your own assignment	❌ No	Academic fraud, misrepresentation of your work
Using AI to write your entire report without disclosure	❌ No	Dishonest — defeats the educational purpose
Asking AI to solve exam questions during an exam	❌ No	Cheating — equivalent to using unauthorized aids
Using AI to write code you don't understand and submitting as yours	❌ No	You don't learn, and it misrepresents your skill

The KMC standard is: AI is your research assistant, your brainstorming partner, and your writing coach — but the final thinking, verification, synthesis, and expression must come from YOU.

CHAPTER 2: Understanding Large Language Models (LLMs)

"A language model is a probability distribution over sequences of words. A large language model is that idea scaled to an almost incomprehensible degree." — Computer Science Perspective

2.1 What Is a Language Model?

A Language Model is an AI system trained to understand and generate human language. At its core, it learns to predict: given the words so far in a sequence, what is the most likely next word? This seemingly simple task, when scaled massively, produces extraordinarily capable systems.

Early language models could predict the next word in a sentence. Large Language Models (LLMs), trained on hundreds of billions of words from the internet, books, code repositories, and scientific papers, can write essays, explain quantum physics, debug code, compose poetry, and hold nuanced conversations.

2.2 The Transformer Revolution

The modern era of LLMs began in 2017 when Google researchers published a paper titled 'Attention Is All You Need.' This paper introduced the Transformer architecture, which is the foundation of every major AI language model today — GPT-4, Gemini, Claude, Llama, and all others.

Key Innovation: The Attention Mechanism

Before Transformers, AI models processed text sequentially — word by word — and had trouble 'remembering' context from earlier in a long document. The Attention Mechanism allows the model to look at ALL words in the input simultaneously and determine which words are most relevant to each other.

🔑 ATTENTION EXAMPLE: How AI Understands Context
`Sentence: "The trophy did not fit in the suitcase because it was too big."`
`What does 'it' refer to — the trophy or the suitcase?`
`→ Humans know instantly: 'it' = the trophy (because the trophy was too big to fit)`

```
Attention mechanism assigns high attention weights between 'it' and
'trophy'
based on the logical relationship 'did not fit' → 'too big'

This contextual understanding is what makes LLMs feel human-like in
conversation.
```

2.3 How LLMs Are Trained: The Three Phases

Phase 1: Pre-training

The model is exposed to massive amounts of text data — essentially a snapshot of human written knowledge. It learns statistical patterns: which words follow which, how sentences are structured, how ideas relate. This phase requires enormous computing resources — thousands of specialized GPUs running for months.

LLM	Estimated Training Data	Parameters	Training Cost (Approx.)
GPT-3	570 GB text data	175 billion	~$4.6 million
GPT-4	~13 trillion tokens	~1.7 trillion (est.)	~$100 million
Claude 3	Undisclosed	Undisclosed	Undisclosed
Gemini Ultra	Multimodal corpus	Undisclosed	Very large
Llama 3 70B	15 trillion tokens	70 billion	Open source

Phase 2: Instruction Fine-tuning (IFT)

Raw pre-trained models produce completions but don't respond to instructions well. Fine-tuning on human-written question-answer pairs teaches the model to follow instructions, answer questions, and have conversations. This is what transforms a text predictor into a helpful assistant.

Phase 3: Reinforcement Learning from Human Feedback (RLHF)

Human raters compare different model outputs and rate which responses are more helpful, accurate, and harmless. The model is then trained to produce outputs that humans prefer. This phase makes the difference between a technically capable but unreliable AI and a genuinely useful assistant.

2.4 Tokens: The Building Blocks of LLM Communication

LLMs don't read text character by character or word by word — they process tokens. A token is roughly a word-piece: common words are usually one token, rare words might be split into multiple tokens, and even punctuation counts as a token.

🔢 TOKEN EXAMPLES (approximately)
"Hello" = 1 token
"Kathmandu" = 2-3 tokens (Kath + man + du)
"AI" = 1 token
"photosynthesis" = 3-4 tokens
"I am a student at KMC" = 6 tokens (roughly)
Rule of thumb: 1 token ≈ 0.75 words in English
100 tokens ≈ 75 words ≈ one short paragraph
1,000 tokens ≈ 750 words ≈ one page of text

Understanding tokens matters because: (1) AI models have a 'context window' — a maximum number of tokens they can process at once, and (2) API pricing is based on tokens consumed. The more tokens in your prompt and response, the more it costs.

Model	Context Window	Approx. Pages of Text
GPT-3.5 Turbo	16,385 tokens	~12 pages
GPT-4o	128,000 tokens	~96 pages
Claude 3.5 Sonnet	200,000 tokens	~150 pages
Gemini 1.5 Pro	1,000,000 tokens	~750 pages
Gemini 2.0 Flash	1,000,000 tokens	~750 pages

2.5 Temperature & Sampling Parameters

When you interact with an LLM, the model doesn't always give the same response to the same prompt. This is because of randomness parameters that control how creative or deterministic the model is.

Temperature

Temperature controls the randomness of responses on a scale from 0.0 to 2.0. A low temperature makes the model more conservative and predictable. A high temperature makes it more creative and surprising — but also more likely to make errors.

Temperature	Behavior	Best For
0.0	Completely deterministic — always picks the most likely token	Factual Q&A, code generation, data extraction
0.3 – 0.5	Mostly predictable with slight variation	Business writing, summarization, technical docs
0.7 – 0.9	Balanced creativity and coherence (DEFAULT)	General use, brainstorming, conversation
1.2 – 1.5	More creative, more unexpected connections	Creative writing, poetry, ideation
2.0	Very random — may produce incoherent output	Experimental artistic generation

2.6 The Major LLMs Compared

Model	Creator	Strengths	Weaknesses	Best Use Case
GPT-4o	OpenAI	Versatile, multimodal, huge ecosystem	Can be verbose, expensive API	General use, coding, images
Claude 3.5 Sonnet	Anthropic	Long context, safety, nuanced reasoning	Fewer plugins/integrations	Long documents, analysis
Gemini 1.5 Pro	Google	1M context, Google Workspace integration	Slightly slower for complex tasks	Research, Google ecosystem
Llama 3.1 70B	Meta	Open source, privacy, free to self-host	Requires technical setup	Privacy-sensitive work, research
Mistral Large	Mistral AI	Fast, efficient, European privacy laws	Smaller ecosystem	Enterprise with GDPR compliance

2.7 Multimodal AI: Beyond Text

The latest generation of LLMs are not limited to text. They can process and generate multiple modalities — text, images, audio, video, and code. This multimodal capability dramatically expands what you can do with AI.

What Multimodal AI Can Do

- Image Analysis: Upload a photo and ask 'What is wrong with this circuit diagram?' or 'Describe this Nepali meal and estimate its nutritional content'

- Document Processing: Upload a PDF business report and ask 'Summarize the key financial risks mentioned in this report'
- Chart Interpretation: Upload a graph and ask 'What trend does this chart show for Q3?'
- Code Screenshot Analysis: Paste a screenshot of code with an error and ask 'What is causing this bug?'
- Medical Image Analysis (Experimental): Some systems can analyze X-rays or medical scans
- Handwritten Notes: Upload a photo of your handwritten notes and ask it to convert to digital text

2.8 The AI Safety Challenge

Every major AI company invests heavily in AI safety — making sure their models don't produce harmful, biased, or dangerous outputs. Understanding these safety measures helps you understand why AI sometimes refuses requests or gives overly cautious responses.

Common Safety Measures

9. Content Filtering: Trained to refuse requests for harmful content — weapons instructions, explicit material, harassment
10. Bias Mitigation: Trained to avoid perpetuating harmful stereotypes about gender, race, nationality, religion
11. Constitutional AI: Claude specifically is trained using 'AI principles' to reason about its own outputs
12. RLHF Safety: Human raters flag harmful outputs during training, reducing their probability
13. Red-teaming: Teams of researchers actively try to break AI safety measures to find vulnerabilities

Important: If an AI refuses your request, it's often because your phrasing triggered a safety filter, not because your intent was harmful. Rephrasing your prompt more specifically and professionally usually resolves this.

CHAPTER 3: The Art and Science of Prompt Engineering

"Prompt engineering is the new programming. The ability to communicate precisely with AI systems will be as valuable as coding skills in the next decade." — AI Research Community

3.1 What Is Prompt Engineering?

Prompt engineering is the practice of designing, testing, and refining the inputs (prompts) you give to AI systems to get the best possible outputs. Think of it as the skill of asking the right questions in the right way.

A prompt is any text input you provide to an AI model. It can be a simple question, a detailed instruction, an example of what you want, a persona you want the AI to adopt, or all of these combined. The quality of your prompt directly determines the quality of the AI's response — this is so reliable that engineers say: 'garbage in, garbage out.'

Prompt engineering is not about tricking AI — it's about communicating with precision. The more context, constraints, and clarity you provide, the more useful and accurate the AI's response will be. A great prompt engineer thinks like a teacher explaining a task to a very capable but very literal student.

3.2 The Anatomy of an Excellent Prompt

The best prompts typically contain some or all of these components. Not every prompt needs all components, but complex tasks benefit from including most of them.

Component	What It Does	Example
Role / Persona	Tells AI who it is	"Act as an experienced marketing consultant with 10 years in Nepal's FMCG sector"
Context	Provides background information	"I am a BBA student at KMC preparing a business plan for a food delivery startup"
Task	Specifies exactly what you want	"Write a SWOT analysis of launching a food delivery app in Kathmandu"
Format	Specifies how the output should look	"Format as a table with 4 quadrants. Use bullet points. Maximum 5 bullets per section"
Constraints	Sets boundaries and limitations	"Focus only on 2024–2025 market conditions. Avoid generic advice. Be specific to Nepal"
Tone	Sets the style of writing	"Professional but accessible. Avoid jargon. Write as if for a presentation to investors"
Examples	Shows what you want (few-shot)	"Example of the quality I need: [paste example output]"

Output Length	Controls verbosity	"Keep total response under 500 words. Be concise and actionable."

3.3 The Role-Context-Task-Constraint (RCTC) Framework

For this workshop, we use the RCTC framework as our primary prompt structure. This is a simplified version of professional prompt engineering that is easy to remember and apply consistently.

🎯 THE RCTC PROMPT FRAMEWORK

```
ROLE:       Define who the AI should be
CONTEXT:    Provide all relevant background information
TASK:       State exactly what you want the AI to do
CONSTRAINT: Set limits, format, length, style requirements

Template: 'You are [ROLE]. I am [who you are] and [CONTEXT].
Your task is to [TASK]. Please [CONSTRAINT].'
```

RCTC in Action — Tourism Promotion Example

❌ BAD PROMPT (0 structure):

```
Tell me about tourism in Nepal.
```

✅ EXCELLENT PROMPT (Full RCTC):

```
ROLE: You are a tourism marketing strategist with expertise in South
Asian travel markets
and digital marketing for the hospitality sector.

CONTEXT: I am a BBA student at Kathmandu Model College working on a
research project
```

```
about post-COVID tourism recovery in Nepal (2023-2025). Nepal received
approximately

1.1 million tourists in 2023, down from the 2019 peak.

TASK: Analyze the top 3 digital marketing strategies that Nepal's
tourism industry

should adopt to attract more tourists from India and China in
2025-2026.

CONSTRAINT: Format as a numbered list with each strategy including: (1)
strategy name,

(2) why it works for Nepal specifically, (3) one concrete
implementation example,

(4) estimated cost range. Keep total response under 600 words. Use
professional

business language suitable for an academic report.
```

3.4 Prompt Patterns: The Professional Toolkit

Pattern 1: Zero-Shot Prompting

You give the AI a task with no examples. Works well for common, well-defined tasks.

ZERO-SHOT EXAMPLE:

```
Translate the following sentence to Nepali: 'Our company provides
excellent customer service.'
```

Pattern 2: Few-Shot Prompting

You provide 2-5 examples of the input-output pattern you want before giving your actual task. Dramatically improves output quality for specialized or unusual formats.

FEW-SHOT EXAMPLE:

Classify the following customer reviews as POSITIVE, NEGATIVE, or NEUTRAL.
Example 1: "The product arrived on time and worked perfectly." → POSITIVE
Example 2: "Terrible quality, broke after one week." → NEGATIVE
Example 3: "The package was average. Nothing special." → NEUTRAL
Now classify these reviews:
1. "Loved the design but the battery life was disappointing."
2. "Super fast delivery, exceeded my expectations!"
3. "It works, I guess."

Pattern 3: Chain-of-Thought (CoT) Prompting

You instruct the AI to 'think step by step' before giving an answer. This dramatically improves reasoning accuracy for math, logic, and complex analysis tasks.

CHAIN-OF-THOUGHT EXAMPLE:
Solve this business problem step by step:
A Kathmandu restaurant has fixed costs of NPR 150,000/month (rent, salaries),
variable costs of 40% of revenue, and wants to make NPR 50,000 net profit.
What minimum monthly revenue do they need? Show all calculation steps.

Pattern 4: Role-Playing Prompting

You assign a specific expert persona to the AI. This activates the model's knowledge in that domain more effectively than a generic request.

ROLE-PLAYING EXAMPLE:
Act as Warren Buffett's investment philosophy AI advisor. Using Buffett's core principles
of value investing, analyze whether a Nepali startup called 'GharKhana' (a local food
delivery app with NPR 2 crore in monthly revenue and 15% month-over-month growth)
would be a good investment. Reference Buffett's actual quotes and principles.

Pattern 5: Self-Consistency Prompting

You ask the AI to approach the same problem multiple ways and then synthesize the best answer. Useful for important decisions.

SELF-CONSISTENCY EXAMPLE:
I need to decide whether to launch a mobile app or a website first for my startup.
Please analyze this decision from THREE different expert perspectives:
1. A startup founder who bootstrapped from Nepal
2. A venture capitalist evaluating potential for investment
3. A UX designer focused on user adoption
After presenting all three perspectives, give me a final recommendation that
synthesizes the most important considerations from all three viewpoints.

3.5 The Iterative Prompt Refinement Process

Professional prompt engineers never expect the first prompt to be perfect. They use an iterative refinement process — write a prompt, evaluate the output, identify weaknesses, refine the prompt, and repeat.

Iteration	Prompt Version	Problem Identified	Refinement Made
1st Try	Write a business plan for a tea shop	Too generic, no Nepal context, missing sections	Add location, budget, target market
2nd Try	Write a business plan for a tea shop in Thamel with NPR 500k budget for tourists	Missing format structure, too short	Specify sections, add format requirements
3rd Try	Full RCTC prompt with all context	Hallucinated some tourism statistics	Add: 'Use only facts you are confident about; flag any uncertain data'
4th Try	Full RCTC + uncertainty instruction	Output is good but too long	Add word limit constraint
Final	Optimized full prompt	Output meets requirements ✅	Save as template for reuse

3.6 Common Prompt Engineering Mistakes

Mistake 1: The Vague Command

❌ BAD — Vague Command	✅ GOOD — Specific Command
Improve my essay.	Review my essay's introduction paragraph for logical flow, clarity, and grammar. Provide specific rewrite suggestions for any weak sentences. Do not rewrite the entire essay.

Mistake 2: Missing Context

❌ BAD — No Context	✅ GOOD — Rich Context
What should I do about this problem?	I run a small Kathmandu-based clothing export business. Our main customer (a German fashion brand) is demanding we reduce prices by 15% or they'll switch suppliers. We have NPR 80L in outstanding invoices with them. What negotiation strategies should I use?

Mistake 3: No Format Specification

❌ BAD — No Format	✅ GOOD — Clear Format
List marketing strategies for my restaurant.	List 5 digital marketing strategies for a Kathmandu restaurant targeting young professionals (24-35). Format each strategy as:

	[Strategy Name] | [Platform/Tool] | [Why it works for this target] | [Estimated monthly cost in NPR] | [Expected result in 3 months].

Mistake 4: Leading the AI to a Specific Answer

❌ BAD — Leading Question	✅ GOOD — Open Analysis
Why is opening a coffee shop in Nepal a great business idea?	Provide an objective analysis of the potential opportunities AND challenges of opening a specialty coffee shop in Kathmandu in 2025. Include market size, competition, supplier availability, and realistic ROI expectations.

3.7 Prompt Templates for Academic Students

Template 1: Research Assistant

RESEARCH TEMPLATE
You are an academic research assistant specializing in [YOUR FIELD].
I am a [YEAR] year [PROGRAM] student at Kathmandu Model College.
I need to research [TOPIC] for my [ASSIGNMENT TYPE].
Please help me:
1. Identify the 5 most important research questions about [TOPIC]
2. Suggest credible sources I should look for (academic journals, reports, data)
3. Provide a brief overview of the current state of knowledge on [TOPIC]
4. Highlight any controversial or debated aspects I should address
Note: Flag any information you are uncertain about. I will verify all facts independently.

Template 2: Essay Improvement

ESSAY IMPROVEMENT TEMPLATE
You are an academic writing coach specializing in business and management writing.
I will paste a paragraph from my essay below.
Please evaluate it on: (1) Logical flow and argumentation, (2) Clarity and conciseness,
(3) Academic tone and vocabulary, (4) Grammar and syntax
For each weakness you identify, show the original sentence AND your suggested
improvement side by side. Explain why your version is better.
Do not rewrite the entire paragraph — only fix what needs fixing.
My paragraph: [PASTE YOUR PARAGRAPH HERE]

Template 3: Business Idea Validator

BUSINESS IDEA TEMPLATE
You are a startup mentor and business analyst with expertise in South Asian markets.
I have a business idea I want to validate before investing time in it.
My idea: [DESCRIBE YOUR IDEA IN 2-3 SENTENCES]
Target market: [WHO ARE YOUR CUSTOMERS]
Location: Nepal/Kathmandu

Initial budget: approximately NPR [AMOUNT]
Please provide:
1. STRENGTHS of this idea (what makes it viable)
2. CRITICAL RISKS (what could kill this business)
3. MARKET SIZE ESTIMATE (rough calculation with assumptions shown)
4. TOP 3 competitors or substitutes
5. VERDICT: Viable / Needs modification / High risk — and why
Be honest and critical. I need real feedback, not encouragement.

CHAPTER 4: ChatGPT – Deep Dive & Mastery

ChatGPT reached 1 million users in just 5 days after launch — the fastest adoption of any consumer technology in history. Understanding it deeply is a competitive advantage.

4.1 What Is ChatGPT?

ChatGPT (Chat Generative Pre-trained Transformer) is an AI chatbot developed by OpenAI. It is built on the GPT series of large language models. When you type a message to ChatGPT, it processes your text, predicts the most contextually appropriate response, and generates it token by token in real time.

ChatGPT is not one single product — it is a family of models with different capability tiers. The free tier uses GPT-3.5 Turbo (capable but older), while ChatGPT Plus subscribers get access to GPT-4o, OpenAI's most powerful model, which can process images, audio, and text simultaneously.

4.2 ChatGPT Interface Tour

Interface Element	What It Does	Beginner Tip
New Chat button	Starts a fresh conversation (clears memory)	Use when switching topics to avoid context confusion
Conversation History	Left sidebar shows past chats	Name your chats for easy retrieval later
Message Input Box	Where you type your prompts	You can paste text, upload images (Plus), and add files
Model Selector	Switch between GPT-3.5 and GPT-4o	Use GPT-4o for complex tasks, 3.5 for simple/fast queries
Voice Mode	Talk to ChatGPT via microphone	Great for brainstorming ideas on the go
DALL-E Button	Generate images from text	Available in GPT-4o — very useful for presentations
Custom Instructions	Set your background context permanently	Set once: your name, role, preferences — saves time
GPT Store	Access specialized custom GPTs	Find GPTs built for specific tasks like coding, writing, research
Share Chat	Get a shareable link to your conversation	Useful for submitting AI-assisted work transparently

4.3 ChatGPT Prompting — Bad vs. Good Examples

Example 1: Business Analysis Task

❌ BAD PROMPT	✅ EXCELLENT PROMPT
What is SWOT analysis? Why It Fails: • Gets a generic textbook definition • No application to your situation • Could have Googled this • Zero business value	Act as a senior business analyst. I'm launching a cloud kitchen (delivery-only restaurant) in New Baneshwor, Kathmandu with NPR 8 lakh capital. Menu: Nepali thali + momos. Target: office workers. Create a detailed SWOT analysis with 4-5 points per quadrant. Be specific to 2025 Kathmandu market. Format as a table.

ChatGPT Output Comparison — Bad Prompt Response

ChatGPT Output (BAD PROMPT — Generic & Useless):
SWOT analysis stands for Strengths, Weaknesses, Opportunities, and Threats.
It is a strategic planning tool used by businesses to assess their position.
Strengths are internal positive factors. Weaknesses are internal negative factors.
Opportunities are external positive factors. Threats are external negative factors.
SWOT analysis helps organizations make better strategic decisions by understanding
their current position and the environment they operate in...
[Continues for 3 more paragraphs with zero specificity]

ChatGPT Output (GOOD PROMPT — Specific & Actionable):
SWOT ANALYSIS: Cloud Kitchen 'KMC Thali' — New Baneshwor, Kathmandu (2025)
━━━

• Low overhead vs. dine-in restaurant	• No brand recognition at launch
• Authentic Nepali thali differentiates	• Limited to delivery radius (2-3 km)
• Momos = high demand, low food cost	• Dependency on Foodmandu/Bhoj.nu margins
• Office worker timing = predictable	• Kitchen visibility limited to packaging
lunch rush (12-2 PM)	• Single kitchen = single point of failure
——————————————

• What tone? • Result: A template so generic it's unusable	time offer (20% off), and a clear CTA button. Tone: Warm, celebratory, premium. Max 200 words.

Example 3: Data Analysis Task

❌ BAD PROMPT	✅ EXCELLENT PROMPT
Analyze this data. [pastes a table of numbers] Why It Fails: • No context about what the data is • No guidance on what analysis is needed • No format for the output • AI will make generic observations	You are a data analyst. The table below shows monthly sales data (in NPR thousands) for a Kathmandu electronics retailer across 4 product categories for Jan-Dec 2024. 1. Identify the top 3 trends 2. Calculate YoY growth rate for each category 3. Identify seasonality patterns with reasons 4. Recommend 2 data-driven actions for 2025 Format: bullet points with supporting numbers.

4.4 ChatGPT Advanced Features

Custom GPTs — Specialized AI Assistants

The GPT Store contains thousands of custom AI assistants built by developers for specific tasks. These custom GPTs have specialized instructions, knowledge bases, and sometimes integrations that make them far more effective for niche tasks than the general-purpose ChatGPT.

Custom GPT Name	What It Does	Best For Students
Scholar AI	Finds and explains academic papers	Research for assignments and dissertations
Consensus	Searches scientific research for answers	Evidence-based arguments in essays
FinanceGPT	Financial analysis and modeling	BBA/BBM financial analysis tasks
Code Copilot	Advanced coding assistance	BCA students building projects
Canva	Create designs from text descriptions	Marketing presentations and visuals
PDF AI	Analyze and summarize uploaded PDFs	Quickly processing research papers

Code Interpreter / Advanced Data Analysis

One of ChatGPT Plus's most powerful features is the Advanced Data Analysis capability. You can upload Excel files, CSVs, or PDFs, and ChatGPT will run actual Python code to analyze them, create charts, and extract insights.

CODE INTERPRETER PROMPT EXAMPLE:
I'm uploading a CSV file containing 6 months of sales data for my family's
textile shop in Patan. The columns are: Date, Product, Units_Sold, Revenue_NPR,
Customer_Type (Walk-in/Online), Payment_Method.
Please: 1. Clean any missing or inconsistent data and tell me what you fixed
2. Create a bar chart showing monthly revenue trends
3. Calculate which product category has the highest profit margin
4. Identify our top 5 highest-value customers by total spend
5. Write a 1-page business summary suitable for showing to a bank for a loan

4.5 ChatGPT System Prompt Mastery

The system prompt is a hidden instruction that sets the AI's behavior for the entire conversation. In ChatGPT, you can set this via Custom Instructions (Settings → Custom Instructions). This is extremely powerful for consistent results.

POWERFUL CUSTOM INSTRUCTION SETUP:
ABOUT ME:
I am a BBA student at Kathmandu Model College, Nepal. I am interested in
entrepreneurship, digital marketing, and the Nepali business landscape.
My assignments often require academic writing (APA format) and business analysis.
HOW TO RESPOND:

1. Always be specific to the Nepal/South Asia context when giving business examples
2. When citing statistics, note your uncertainty level and suggest where to verify
3. For business advice, structure responses with clear headings
4. If my prompt is unclear, ask ONE clarifying question before proceeding
5. When writing essays, maintain academic tone but avoid unnecessary jargon

CHAPTER 5: Google Gemini – Deep Dive & Mastery

Gemini is Google's answer to GPT-4 — and in several dimensions, particularly research and its integration with Google's ecosystem, it actually surpasses it.

5.1 What Is Google Gemini?

Google Gemini is a family of multimodal AI models developed by Google DeepMind. 'Multimodal' means it natively understands text, images, audio, video, and code — not as separate add-ons, but as integrated capabilities. The Gemini family includes Gemini Ultra (most capable), Gemini Pro (balanced), and Gemini Nano (runs on mobile devices).

Gemini is available as a standalone product at gemini.google.com, and is also integrated throughout Google's product ecosystem — Google Search, Google Docs, Google Sheets, Gmail, Google Photos, and Android devices. If you use Google Workspace for school or work, Gemini is already embedded in your tools.

5.2 Gemini vs ChatGPT — Key Differences

Dimension	ChatGPT (GPT-4o)	Google Gemini 1.5 Pro
Context Window	128,000 tokens (~96 pages)	1,000,000 tokens (~750 pages)
Real-time Web Access	Available with browsing tool	Native — always searches when relevant
Google Integration	None	Deep — Docs, Sheets, Gmail, Drive
Image Generation	DALL-E 3 (separate model)	Imagen 3 integration
Code Execution	Advanced Data Analysis (sandbox)	Google Colab integration
YouTube Understanding	Cannot process YouTube videos	Can analyze YouTube videos natively
Price (Free Tier)	GPT-3.5 only	Gemini 1.5 Flash free
Multimodal	Text + Image + Audio	Text + Image + Audio + Video + Code
Strength	Plugin ecosystem, broader training	Research, real-time info, Google products

5.3 Gemini's Superpower: Real-Time Research

Unlike ChatGPT's base version, Gemini has native access to Google Search and provides citations for factual claims. This makes it significantly more reliable for research tasks and dramatically reduces hallucination risk on current events.

Bad vs Good Gemini Prompts for Research

❌ BAD RESEARCH PROMPT	✅ EXCELLENT RESEARCH PROMPT
What's the current economic situation in Nepal? Problems: • Too broad — what aspect? • 'Current' with no date context • No guidance on what to include • Will produce a generic overview • No request for sources	Search for the latest data (2024-2025) on Nepal's economic situation. I need specifically: 1. GDP growth rate (official NRB/CBS data) 2. Remittance inflow as % of GDP 3. Foreign Direct Investment (FDI) trends 4. Tourism sector recovery statistics For each point, provide: the specific number, the exact source name, and the date published. Flag if any data is older than 6 months.

GEMINI ACTUAL OUTPUT (Good Prompt — with citations):
Based on current search results, here is Nepal's economic data:
1. GDP Growth Rate: Nepal's economy grew by 3.9% in FY 2023/24
Source: Nepal Rastra Bank (NRB) Annual Report, October 2024
2. Remittance: Nepal received remittances equivalent to 26.1% of GDP in 2023/24
— among the highest rates in South Asia
Source: World Bank Migration and Development Brief, 2024
3. FDI: Net FDI inflow was NPR 6.78 billion in 2023/24, a recovery from 2022/23
Source: Department of Industry, Nepal — Annual Report 2024
4. Tourism: Nepal received 1,048,669 tourists in 2023, with India (21.5%),
USA (13.2%), and China (4.3%) as top source markets

Source: Nepal Tourism Statistics 2023, NTB
⚠️ Note: FY 2024/25 mid-year data not yet published. Above figures are most current available.

5.4 Gemini for Document Analysis

One of Gemini's most practical features for students is its ability to process extremely long documents — up to the equivalent of 750 pages in a single prompt. This makes it invaluable for analyzing annual reports, textbooks, research papers, or lengthy business documents.

GEMINI DOCUMENT ANALYSIS PROMPTS:
PROMPT 1 — Annual Report Analysis:
[Upload PDF of company annual report]
Analyze this annual report and extract: (1) Key financial ratios with year-over-year
comparison, (2) Major strategic priorities for next year, (3) Risk factors mentioned,
(4) CEO's main message in 3 bullet points. Format as an executive briefing.
PROMPT 2 — Research Paper Summary:
[Upload research paper PDF]
Summarize this academic paper for a BBA student audience. Include: research question,
methodology, key findings, limitations, and implications for business practice in Nepal.
Use simple language — avoid excessive academic jargon.
PROMPT 3 — Contract Review:

[Upload business contract PDF]
Review this contract and flag: (1) Any unusual clauses, (2) Payment terms and deadlines,
(3) Termination conditions, (4) Intellectual property provisions, (5) Liability clauses.
Note: This is educational review only, not legal advice.

5.5 Gemini for Google Workspace

Gemini in Google Docs

With Google Workspace (available free with your college email), Gemini is embedded directly in Google Docs. You can use the 'Help me write' feature to generate content, or ask Gemini to improve, summarize, or analyze your documents without leaving the application.

Task in Google Docs	Gemini Prompt	Time Saved
Draft an essay introduction	@Gemini Help me write an introduction for an essay about digital transformation in Nepal's banking sector	20-30 minutes
Improve a paragraph	Select text → Ask Gemini → 'Make this more formal and concise'	5-10 minutes
Create an outline	@Gemini Create a structured outline for a business plan for a Kathmandu-based tech startup	15-20 minutes
Add citations	@Gemini Find 3 academic sources that support this argument about FDI in Nepal	30-45 minutes

5.6 Gemini Multimodal Prompting

Analyzing Images with Gemini

Gemini can analyze images with remarkable sophistication. You can upload photos, diagrams, charts, or screenshots and ask contextual questions about them.

IMAGE ANALYSIS PROMPT EXAMPLES:
EXAMPLE 1 — Market Research Photo:
[Upload photo of a street market in Thamel]

Analyze this photo of a Kathmandu street market. Identify: types of products being sold,
estimated customer demographics visible, quality of the retail environment,
and suggest 3 improvements that would increase sales based on retail design principles.
EXAMPLE 2 — Financial Chart:
[Upload chart from annual report]
Analyze this financial chart. What trend is shown? Calculate the approximate
percentage change between the highest and lowest points. What business event
might explain the spike visible in Q3?
EXAMPLE 3 — Your Own Handwritten Notes:
[Photo of handwritten notes]
Convert these handwritten notes to clean digital text, organized with proper
headings and bullet points. If anything is illegible, indicate with [?].

CHAPTER 6: Claude by Anthropic – Deep Dive & Mastery

'Claude is designed to be helpful, harmless, and honest — a combination that makes it uniquely trustworthy for academic and professional work.' — Anthropic

6.1 What Makes Claude Different?

Claude is an AI assistant developed by Anthropic, a company founded by former OpenAI researchers with a specific focus on AI safety. Claude's defining characteristics are its long context window (up to 200,000 tokens), its tendency for nuanced, thoughtful reasoning, and its 'Constitutional AI' training approach which makes it particularly transparent about uncertainty and limitations.

Claude is widely regarded by researchers and professionals as the best AI for tasks that require: careful long-form reasoning, analysis of lengthy documents, nuanced ethical considerations, and highly polished writing. Many professional writers, lawyers, and researchers prefer Claude for exactly these reasons.

6.2 Claude's Unique Strengths

Strength	Why It Matters	Use Case Example
200K Token Context	Can process entire books, legal files, large codebases	Analyzing a full-length annual report or dissertation
Honesty About Uncertainty	Tells you when it's unsure instead of confidently hallucinating	Research where accurate information is critical
Long-Form Writing	Maintains coherent style, argument, and tone across long texts	Writing a full business report or academic paper
Constitutional AI	Trained to reason about ethics transparently	Sensitive topics requiring balanced perspective
Code Quality	Writes clean, documented, production-quality code	BCA projects and software development tasks
Instruction Following	Very precise — follows complex multi-part instructions	Complex structured outputs with specific format requirements

6.3 Bad vs Good Claude Prompts — Extended Examples

Example 1: Long-Form Business Report

❌ WEAK PROMPT	✅ POWERFUL PROMPT
Write a report about e-commerce in Nepal. What you get: • A generic 500-word overview • No structure or sections • Generic statistics • No actionable insights • Could apply to any country	You are a senior business consultant. Write a structured 1200-word analytical report titled: 'E-Commerce in Nepal: Opportunities and Barriers for Local Entrepreneurs (2024-2026)' Sections: Executive Summary, Market Overview, Key Players (Daraz, SastoDeal, Bhoj.nu etc.), Barriers (logistics, digital payments, trust), 3 Strategic Recommendations for new entrants. Cite data with source names. Professional tone.

Example 2: Code Generation (for BCA Students)

❌ WEAK CODE PROMPT	✅ EXCELLENT CODE PROMPT
Write Python code to analyze sales data. What you get: • Generic code with placeholder variables • No comments or documentation • May not match your data format • Basic functionality only • No error handling	Write production-quality Python code to analyze a CSV file named 'sales_2024.csv' with columns: Date (YYYY-MM-DD), Product, Category, Units, Revenue_NPR, Customer_ID. Requirements: Load with pandas, handle missing values, calculate monthly revenue by category, plot a stacked bar chart with matplotlib, export summary to 'report.xlsx'. Add docstrings and error handling for file not found.

CLAUDE'S CODE OUTPUT (Good Prompt):

```
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
from pathlib import Path
import sys

def load_and_validate(filepath: str) -> pd.DataFrame:
    """Load sales CSV and validate required columns."""
    path = Path(filepath)
```

```
    if not path.exists():
        raise FileNotFoundError(f"Sales file not found: {filepath}")
    df = pd.read_csv(filepath, parse_dates=['Date'])
    required =
{'Date','Product','Category','Units','Revenue_NPR','Customer_ID'}
    missing = required - set(df.columns)
    if missing:
        raise ValueError(f"Missing columns: {missing}")
    df = df.dropna(subset=['Date','Revenue_NPR'])
    print(f'Loaded {len(df)} records, dropped
{df.Revenue_NPR.isna().sum()} null rows')
    return df

def monthly_category_revenue(df: pd.DataFrame) -> pd.DataFrame:
    """Compute monthly revenue pivot by category."""
    df['Month'] = df['Date'].dt.to_period('M')
    pivot =
df.groupby(['Month','Category'])['Revenue_NPR'].sum().unstack(fill_valu
e=0)
    return pivot

# [Full implementation continues...]
```

6.4 Claude for Academic Research — Long Context Power

Claude's 200,000 token context window is its most distinctive advantage. Here are practical ways to leverage this for academic work:

ADVANCED ACADEMIC USE CASES:
USE CASE 1: Full Dissertation Review
Paste your entire 15,000-word dissertation draft and ask:
'Review this full dissertation draft. Identify: (1) logical inconsistencies between chapters,
(2) unsupported claims that need citations, (3) repetitive sections that should be condensed,
(4) arguments in the literature review that contradict my research findings.
Provide specific chapter and section references for each issue.'
USE CASE 2: Comparative Analysis Across Multiple Documents
Paste 3 different company annual reports and ask:
'Compare these three annual reports and identify: (1) which company has the strongest
financial position and why, (2) common risks all three companies mention,
(3) contradictions in their stated strategies vs. financial performance.'
USE CASE 3: Iterative Essay Improvement
Start a Claude conversation with your essay draft. Then in subsequent messages:
- 'Improve the logical flow between paragraphs 2 and 3'
- 'Make the introduction more engaging without losing academic tone'
- 'Check if my conclusion adequately addresses all points raised in the introduction'

6.5 Claude vs ChatGPT vs Gemini — When to Use Which

Task	Best Tool	Why
Quick factual research with sources	Gemini	Real-time search with citations
Long essay writing (1000+ words)	Claude	Best coherence over long texts
Coding with complex requirements	Claude or GPT-4o	Both excellent; Claude for documentation
Image analysis and visual tasks	GPT-4o or Gemini	Both support DALL-E 3 / Imagen 3
Current events and news analysis	Gemini or Perplexity	Real-time web access
Google Docs/Sheets integration	Gemini	Native Workspace integration
Brainstorming and creative ideas	ChatGPT	More playful and divergent
Sensitive analysis (ethics, bias)	Claude	More nuanced and honest
Data analysis with uploaded files	ChatGPT (Code Interpreter)	Best Python execution environment
Academic writing with verification	Perplexity + Claude	Find sources, then write

CHAPTER 7: Gamma.app – AI Presentation Engineering

'What used to take 3 hours in PowerPoint now takes 3 minutes in Gamma. But the best results still come from excellent prompting.' — Product Design Community

7.1 What Is Gamma.app?

Gamma.app is an AI-powered presentation and document creation tool. Unlike traditional PowerPoint where you start with a blank slide and add content, Gamma starts from your text description or outline and generates a complete, professionally designed presentation in seconds.

Gamma creates presentations that are web-native — they can be viewed in any browser, shared via link, embedded in websites, and even have interactive elements. They can also be exported as PowerPoint files or PDFs for traditional distribution.

7.2 Gamma Workflow for Students

Step	Action	Best Practice
1. Create New	Click 'New AI' → Choose 'Presentation'	Start with a clear topic in mind
2. Prompt Input	Type your presentation description	Use the full prompt template below
3. Outline Review	Gamma shows an outline — edit before generating	ALWAYS review and customize the outline
4. Generation	AI creates all slides simultaneously	Choose a theme that matches your brand/college
5. Edit Slides	Click any element to edit text, images, layout	Replace AI images with real ones where possible
6. Add Images	Use the image search or upload your own	Search for Nepal-specific images

7. Export/Share	Share link, export as PPT or PDF	Test the link before presenting

7.3 Gamma Prompting — Bad vs Good

Example 1: Startup Pitch

❌ BAD GAMMA PROMPT	✅ EXCELLENT GAMMA PROMPT
Make a presentation about my food app. Result: • 5 generic slides • No brand identity • Stock photos irrelevant to Nepal • Generic business advice slides • Looks like every other student deck	Create a 10-slide investor pitch deck for 'DhokaKhana' — an AI-powered home-cooked food marketplace connecting Kathmandu housewives with office workers. Slides needed: Cover, Problem, Solution, Market Size (Nepal food delivery market), Business Model, Traction/Metrics, Team, Roadmap, Financials, Ask (NPR 50L seed). Tone: Professional startup pitch. Color: Deep teal and gold. Include real metrics if possible.

Example 2: Academic Presentation

❌ BAD GAMMA PROMPT	✅ EXCELLENT GAMMA PROMPT
Make a slide show about digital marketing. Result: • Generic marketing textbook slides • No Nepal context • Too many text-heavy slides • No clear thesis or argument • Doesn't connect to any coursework	Create an 8-slide academic presentation titled: 'Digital Marketing Adoption Among SMEs in Kathmandu Valley: Challenges and Opportunities' For: BBA 3rd Year Presentation, KMC Slides: Title, Executive Summary, Research Context (Nepal SME landscape), Key Findings (3 slides), Recommendations, Conclusion. Style: Clean academic. Data visualization on findings slides. Citation format: APA. Dark navy and white color scheme.

7.4 Gamma Prompt Templates

GAMMA MASTER PROMPT TEMPLATE:

```
Create a [NUMBER]-slide [TYPE: pitch deck/presentation/report/proposal]
titled:
```

'[YOUR TITLE]'
PURPOSE: [e.g., investor pitch / academic presentation / business proposal / sales deck]
AUDIENCE: [e.g., investors / professors / classmates / clients]
CONTEXT: [2-3 sentences about your topic and situation]
REQUIRED SLIDES (in order):
1. [Slide 1 title and what it should contain]
2. [Slide 2 title and content]
... [continue for each slide]
DESIGN PREFERENCES:
- Color scheme: [e.g., deep blue and white / warm terracotta / corporate navy and gold]
- Style: [professional / academic / startup / creative]
- Include data visualizations on: [specify which slides]
- Font style: [modern / traditional / bold / elegant]
TONE: [formal / conversational / inspirational / analytical]

7.5 Gamma for the Mini-Startup Innovation Activity

During Day 2 of your workshop, you will use Gamma to create a 3-slide pitch for your mini-startup. Here's the optimized approach:

Slide	Required Content	Gamma Prompt Section

Slide 1: The Problem	What problem are you solving? How big is it? Who suffers from it?	Problem: [describe with a hook statistic and relatable scenario]
Slide 2: Our Solution	Your startup name, logo, core product/service, key differentiator	Solution: [startup name] solves this by [description]. Include logo placeholder.
Slide 3: Why Us / Next Steps	Your team's unique strengths, immediate next steps, what you're asking for	Call to Action: [what you want from the audience]

WORKSHOP MINI-STARTUP PITCH PROMPT (Copy & Customize):
`Create a 3-slide startup pitch for our mini-startup called '[YOUR STARTUP NAME]'.`
`SLIDE 1 - THE PROBLEM:`
`Problem: [describe the problem your startup solves, e.g., traffic congestion,`
`tourism information gap, local business digitalization]`
`Who it affects: [target customer group]`
`Scale: [how big is this problem in Kathmandu/Nepal]`
`SLIDE 2 - OUR SOLUTION:`
`Product name: [your startup name]`
`What it does: [one sentence description]`
`How it uses AI: [specific AI feature or tool you'll use]`
`Key benefit: [primary value to customer]`
`SLIDE 3 - OUR TEAM & ASK:`
`Team: 5 KMC students with skills in [list team members' skills]`

What we need: Feedback, mentorship, and pilot customers
Design: Bold, modern, startup style. Use [YOUR TEAM'S CHOSEN COLOR PALETTE].

CHAPTER 8: Leonardo.ai – AI Visual Generation Mastery

'A picture is worth 1,000 words — and with AI image generation, it now takes about 10 words to create a picture worth a thousand.' — Design Community

8.1 What Is Leonardo.ai?

Leonardo.ai is an AI image generation platform that uses Stable Diffusion and custom-trained models to generate high-quality images from text prompts. It is particularly popular for creating logos, marketing visuals, product mockups, concept art, and character designs.

Leonardo offers significant free tier usage (150 tokens/day, renewing daily), making it accessible for student projects. Unlike Midjourney which requires a Discord subscription, or DALL-E which requires ChatGPT Plus, Leonardo's free tier is genuinely usable for creating professional-quality visuals for your workshop projects.

8.2 Leonardo Interface Overview

Feature	What It Does	How to Use
Image Generation	Main text-to-image generation	Go to 'AI Image Generation' → Enter prompt → Generate
Model Selection	Choose the AI model for your style	Recommended: Leonardo Diffusion XL or PhotoReal v2
Guidance Scale	How closely AI follows your prompt (1-20)	7-12 for most uses; higher = more literal
Image Dimensions	Set width and height	1:1 for logos, 16:9 for presentations, 9:16 for social
Negative Prompt	Tell AI what to EXCLUDE	Always use: 'blurry, low quality, distorted, watermark'
PhotoReal Mode	Photorealistic image generation	Best for realistic product mockups and people
Alchemy Mode	Enhanced quality (costs more tokens)	Use for final versions of important visuals
Canvas Editor	Edit generated images with AI tools	Remove backgrounds, add elements, upscale

8.3 The Leonardo Prompt Formula

Image generation prompts follow a different structure than text AI prompts. The formula is:

LEONARDO PROMPT FORMULA:
[SUBJECT] + [STYLE] + [MOOD/LIGHTING] + [TECHNICAL SPECS] + [NEGATIVE PROMPT]
SUBJECT: What the image shows (main object, scene, character)
STYLE: Art style or rendering style (realistic, minimalist, flat design, watercolor)
MOOD/LIGHTING: Atmosphere and lighting conditions (cinematic, warm golden hour, neon)
TECHNICAL SPECS: Composition, camera angle, quality descriptors
NEGATIVE PROMPT: What to exclude (blurry, text, watermark, distorted faces)

8.4 Leonardo Prompting — Bad vs Good Examples

Example 1: Company Logo

❌ BAD IMAGE PROMPT	✅ EXCELLENT IMAGE PROMPT
Logo for my food delivery company Result: • Extremely generic logo • No brand personality • Random colors • May include text that's distorted • Cannot be used professionally • Looks like clip art	Minimalist vector logo design for 'DhokaKhana' a Nepali home-food delivery startup, featuring a traditional Newari wooden window (tikijhya) merged with a modern food delivery scooter silhouette, flat design style, warm terracotta and deep teal color palette, clean lines, professional brand identity, white background, SVG style ---NEGATIVE: text, words, letters, realistic photographic, cluttered, low quality, shadows

Example 2: Marketing Visual for Social Media

❌ BAD PROMPT	✅ EXCELLENT PROMPT
Nepali restaurant photo for Instagram Problems: • No composition guidance	Professional food photography, steaming dal bhat thali plate with gundruk, achar, and ghee rice, served on a traditional beaten copper plate, warm atmospheric lighting, rustic wooden table,

• No style specification • No color direction • Could be anything from a dive bar to fine dining • Unprofessional result	shallow depth of field, bokeh background showing Himalayan decor, overhead 45-degree shot, National Geographic food photography style, ultra detailed NEGATIVE: cartoon, illustration, western food, text overlay, blurry, oversaturated

Example 3: Startup Concept Art

STARTUP VISUALIZATION PROMPT (Tourism Tech App):
`Isometric 3D illustration of a smart tourism app interface for Nepal,`
`showing Mount Everest as a 3D icon in the center, surrounded by floating UI cards`
`displaying: trekking routes, hotel bookings, local guide profiles, and weather forecasts,`
`modern tech startup aesthetic, flat isometric style, vibrant colors: deep blue, white,`
`and himalayan orange, clean professional app mockup, dribbble worthy design,`
`no text or letters, high detail, digital art`
`---NEGATIVE: photorealistic, blurry, text, words, numbers, watermark, dark background`

8.5 Common Leonardo.ai Prompting Mistakes

Mistake	Problem	Solution
Asking for text in images	AI cannot reliably render legible text in images — it distorts letters	Generate without text; add text separately in Canva or PowerPoint
Too many concepts in one prompt	AI gets confused and produces a chaotic result	Focus on ONE main concept with supporting details
No style specification	Random inconsistent art style	Always specify: realistic, flat design, watercolor, 3D render, etc.
No negative prompt	Unwanted elements (extra fingers, blurry bg, watermarks) appear	Always add: 'blurry, low quality, distorted, watermark, extra limbs'

Too short a prompt	Vague output missing key details	More specific = more accurate; aim for 40-80 words
Wrong model for the task	PhotoReal for logos (wrong) or Diffusion for photos (wrong)	Match model to use case: PhotoReal for photos, Leonardo for logos

8.6 Workshop Logo Creation Workflow

During the workshop Mini-Startup activity, your Creative Designer (Role 3) will create a logo for your startup using Leonardo.ai. Follow this workflow:

14. Brainstorm: As a group, agree on 3 design directions — different color palettes, symbols, or styles
15. Generate 3 Variations: Use the prompt template below, changing key elements for each direction
16. Vote & Select: Pick the strongest base image as your starting point
17. Refine: Use Canvas Editor to remove background, adjust colors
18. Export: Download as PNG with transparent background for use in Gamma

WORKSHOP LOGO PROMPT TEMPLATE:
`Minimalist professional logo design for '[YOUR STARTUP NAME]',`
`a [one sentence description of startup type],`
`featuring [symbolic element that represents your business],`
`[art style: flat vector / minimal line art / geometric / badge],`
`[color 1] and [color 2] color palette,`
`white or transparent background,`
`clean modern startup aesthetic, scalable icon,`
`professional brand identity design`
`---NEGATIVE: text, words, letters, blurry, realistic photographic,`
`complicated, cluttered, low quality, watermark, multiple logos`

CHAPTER 9: Perplexity AI – Research & Fact Verification

'Perplexity is what Google Search should have become — answers, not links, with sources you can verify.'

9.1 What Is Perplexity AI?

Perplexity AI is an AI-powered search engine that combines the capabilities of a traditional search engine with a language model. Unlike ChatGPT which relies on training data, Perplexity searches the live web, reads current pages, synthesizes the information, and presents a direct answer with cited sources — all in real time.

For students and researchers, Perplexity solves the #1 problem with using AI for academic work: hallucinations. Because Perplexity shows you exactly which sources it used, you can click through and verify every claim independently. This makes it the safest AI tool for factual research.

9.2 Perplexity vs Google Search vs ChatGPT

Feature	Google Search	ChatGPT	Perplexity AI
Answers directly?	No — shows links	Yes — but may hallucinate	Yes — with citations
Sources provided?	Links to websites	No citations by default	Yes — numbered citations
Real-time web access?	Yes	Limited (with Browse)	Yes — always
Synthesizes info?	No — you must read links	Yes	Yes
Risk of hallucination?	N/A (shows raw pages)	High without verification	Lower — sources cited
Best for?	Finding specific pages	Creative tasks, writing	Academic research, fact-checking
Free tier?	Free	Free (GPT-3.5)	Free (limited searches)

9.3 Perplexity Prompting for Research

Research Workflow: From Question to Verified Answer

PERPLEXITY RESEARCH WORKFLOW:

STEP 1 — Broad Discovery Question:
Ask a high-level question to understand the landscape
Example: "What is the current state of fintech adoption in Nepal in 2024-2025?"
STEP 2 — Deep Dive Questions:
Follow up with specific questions based on interesting points from Step 1
Example: "What percentage of Nepali adults have mobile banking accounts? Cite official sources."
STEP 3 — Source Verification:
Click every citation number to verify the source exists and says what Perplexity claims
Example: Click [1], [2], [3] beside each claim to read the original source
STEP 4 — Counter-Evidence Check:
Ask: "Are there studies or reports that contradict this finding? What is the critical view?"
STEP 5 — Synthesis:
Take your verified facts to Claude or ChatGPT to synthesize into a coherent report

9.4 Bad vs Good Perplexity Prompts

❌ BAD PERPLEXITY PROMPT	✅ EXCELLENT PERPLEXITY PROMPT
Tell me about tourism in Nepal.	Find the most recent official data on Nepal's

Problems: • Gets a broad overview • No specific data requested • Citations will be for generic articles • Not useful for research • Same as Googling the topic	tourism statistics for 2023 and 2024. Specifically: 1. Total tourist arrivals by year 2. Top 5 source countries with percentages 3. Average tourist spending per day in USD 4. Trekking permit revenues Prioritize Nepal Tourism Board (NTB) and Nepal Rastra Bank official sources. Flag if any data is from unofficial or secondary sources.

PERPLEXITY OUTPUT (Good Prompt — with real citations structure):
Based on Nepal Tourism Board and official sources:
[1] Nepal received 1,048,669 international tourist arrivals in 2023,
a significant recovery from 616,000 in 2022 post-COVID.
Source: Nepal Tourism Statistics 2023, Nepal Tourism Board
[2] Top 5 source countries (2023):
India: 224,000 (21.4%) | USA: 138,000 (13.2%) | UK: 83,000 (7.9%)
Australia: 60,000 (5.7%) | Germany: 57,000 (5.4%)
Source: Nepal Tourism Board Annual Statistical Report, 2023
[3] Average tourist expenditure: USD 58.5 per day
Average stay: 12.3 days per tourist
Total tourism receipts: USD 753 million (2023)
Source: NRB Economic Report, Quarter 3, 2024
[4] ⚠️ Trekking permit data: 2024 figures not yet published at time of search.

```
2023 data: 61,382 trekking permits issued. Source: TIMS Center
```

CHAPTER 10: Business Case Studies – Bad vs. Good Prompts

This chapter presents comprehensive real-world business scenarios, comparing ineffective prompts with professional-grade prompts across multiple AI tools. Each case study shows actual-style output comparisons.

CASE STUDY 1: Market Research for a New Product

Scenario

You are a BBA student who has been tasked with conducting market research for a startup that wants to launch organic Himalayan honey products targeting urban Nepali consumers and the export market. Your team has 48 hours to produce a professional market analysis.

Phase 1 — Using Perplexity (Research & Data)

❌ AMATEUR APPROACH	✅ PROFESSIONAL APPROACH
Search: 'honey market nepal' What you get: • Random blog articles • Outdated statistics • No citation to official data • Mix of import/export data • Unusable for professional work Time wasted: 2 hours reading irrelevant pages	Perplexity prompt: 'Find official data on Nepal's honey production, export statistics, and organic food market trends for 2022-2024. Sources: Trade Promotion Centre Nepal, Ministry of Agriculture, TEPC export data. Specifically: total honey production (MT), export value (USD), main export markets, and organic certification status.' Time: 20 minutes, verified citations

Phase 2 — Using ChatGPT (Competitive Analysis)

PROFESSIONAL CHATGPT PROMPT — Competitive Analysis:
`You are a market research analyst specializing in Nepal's FMCG and agro-products sector.`
`Context: I am researching the organic honey market in Nepal for a startup planning to`

launch premium Himalayan honey. The product will be priced at NPR 800-1200 per 500g
(2x the average local brand price). Target market: Urban Kathmandu consumers (25-45,
household income > NPR 50,000/month) + export to India and UAE markets.
Task: Conduct a competitive landscape analysis covering:
1. Top 5 existing honey brands sold in Kathmandu (include price points, positioning)
2. Direct competitors (premium/organic honey products)
3. Substitute products that consumers buy instead of honey
4. Key competitive advantages a new 'Himalayan organic' brand could build
5. Potential distribution channels (supermarkets, health stores, online, export agents)
Format: Professional market research report structure with headers.
Flag any estimates with '(estimated)' notation.

Phase 3 — Using Claude (Full Strategy Report)

CLAUDE PROMPT — Synthesized Strategy Report:
You are a business strategy consultant. I will paste market research findings below.
Using this information, write a 1,500-word strategic market entry report for 'HimaliMad'
— a new premium organic Himalayan honey startup in Nepal.

Structure the report as follows:
1. Executive Summary (150 words)
2. Market Opportunity Analysis (300 words) — include market size, growth rate, gaps
3. Target Customer Profiles (200 words) — 2-3 specific personas
4. Positioning Strategy (200 words) — how to differentiate from competitors
5. Go-to-Market Strategy (300 words) — phase 1 launch plan
6. Financial Projections Overview (150 words) — year 1 revenue targets with assumptions
7. Top 3 Risks and Mitigation Strategies (200 words)
Writing style: Professional consulting report. Use data-driven language.
Include specific numbers and actionable recommendations.
Here are the research findings: [PASTE YOUR PERPLEXITY RESEARCH RESULTS HERE]

CASE STUDY 2: Digital Marketing Campaign

Scenario

You are a BBM student helping a family friend who owns 'Patan Handicrafts' — a traditional Nepali handicraft shop in Lalitpur. The shop has been running for 20 years with walk-in customers but has almost zero online presence. The owner has NPR 50,000/month to spend on digital marketing for 3 months.

The Cascade Prompting Approach

Professional use of AI for marketing requires a cascade approach — each prompt builds on the output of the previous one.

PROMPT CASCADE STRUCTURE:
PROMPT 1 (ChatGPT) — Customer Persona Creation:
'Create 3 detailed customer personas for a traditional Nepali handicraft shop in Lalitpur.
Include: name, age, occupation, monthly income, shopping behavior, social media usage,
what motivates them to buy handicrafts, price sensitivity, preferred payment methods.
Nepal market context. Be specific — no generic descriptions.'
PROMPT 2 (Gemini) — Competitor Research:
'Search for top Nepali handicraft brands with online presence in 2024. What social
media platforms do they use? What type of content gets the most engagement?
Find their most popular posts and products. Cite sources.'
PROMPT 3 (ChatGPT, using personas + competitor data) — Content Calendar:
'Using the [PASTE PERSONAS] and [PASTE COMPETITOR INSIGHTS], create a 30-day
Instagram and Facebook content calendar for Patan Handicrafts. Include: post type,
caption draft, hashtags, best posting time, and NPR estimated boost budget per post.
Budget: NPR 50,000 total for 3 months. Format as a table.'
PROMPT 4 (Claude) — Full Campaign Strategy Document:

'Synthesize the personas, competitor analysis, and content calendar into a
professional 3-month digital marketing strategy document for Patan Handicrafts.'

Content Creation with Leonardo.ai

LEONARDO PROMPTS FOR HANDICRAFT MARKETING:
PRODUCT SHOT PROMPT:
Professional product photography style, traditional Nepali Thanka painting hanging
against a textured white lime-washed wall, golden hour side lighting, artistic
composition, shallow depth of field, museum-quality display, clean background,
high-end craft gallery aesthetic, rich warm colors, detailed fabric texture visible
NEGATIVE: blurry, dark background, cluttered, multiple products, text overlay
INSTAGRAM STORY VISUAL PROMPT:
Flat lay composition of traditional Nepali handicrafts — wooden maskwork, pashmina,
singing bowls arranged artfully on a cream linen background, seen from above,
warm autumn color palette, editorial lifestyle photography, minimalist boho aesthetic,
Pinterest-worthy composition, soft natural lighting, professional product photography
NEGATIVE: cartoon, illustration, text, watermark, cluttered, low quality

CASE STUDY 3: Business Plan for Investor Presentation

Scenario

Three BBA students at KMC want to create a tech startup — an app called 'ParkSathi' that uses AI and camera recognition to find and book parking spaces in real time in Kathmandu, where parking is a massive and growing problem. They have a meeting with a small angel investor in 2 weeks.

Document Needed	AI Tool	Prompt Approach
Market Research	Perplexity	Official data on Kathmandu vehicle growth, parking revenue, smart city initiatives
Financial Model	ChatGPT + Code Interpreter	Build Excel-ready financial projections with NPR figures, 3-year P&L
Pitch Deck	Gamma.app	12-slide investor pitch using RCTC framework, startup style
Logo & Branding	Leonardo.ai	Modern tech logo, parking + tech iconography, blue and white palette
Full Business Plan	Claude	15-page business plan synthesizing all research
Investor Email	ChatGPT	Compelling cold email requesting meeting, professional tone

PARKSATHI INVESTOR PITCH — GAMMA PROMPT:
Create a 12-slide investor pitch deck for 'ParkSathi' — an AI-powered smart parking
app for Kathmandu, Nepal.
Slide breakdown:
1. COVER: ParkSathi tagline + bold visual of Kathmandu traffic
2. THE PROBLEM: Kathmandu has 900,000+ registered vehicles, <15% parking infrastructure
3. THE COST: Average driver wastes 23 minutes finding parking; NPR 2M/day lost productivity
4. OUR SOLUTION: ParkSathi app — real-time camera AI detects empty spots, book & pay in-app

5. HOW IT WORKS: Simple 3-step infographic (Find → Book → Pay)
6. MARKET SIZE: TAM = NPR 3.8B, SAM = Kathmandu Valley, SOM = Year 1 target
7. BUSINESS MODEL: Commission (15%) + Monthly parking operator subscription
8. TRACTION: Pilot with 3 Kathmandu malls, 450 beta users, NPR 2.3L Month 3 GMV
9. COMPETITIVE ADVANTAGE: First mover, hardware partnerships, proprietary AI model
10. TEAM: Founder backgrounds (tech + business + local govt network)
11. FINANCIALS: Revenue projections Y1-Y3, path to profitability Q8
12. THE ASK: NPR 1.5 Crore seed for tech team + camera hardware expansion
Style: Bold modern startup, dark navy background with electric blue accents,
data visualizations on slides 6, 7, 8, 11. Professional, confident, investor-grade.

CASE STUDY 4: Academic Research Report

Scenario

A BBM student needs to write a 3,000-word research report on 'The Impact of Remittances on Consumer Behavior in Nepal' for their Macroeconomics course. The professor requires at least 8 academic citations and original analysis.

STEP-BY-STEP AI-ASSISTED RESEARCH PROCESS:
STEP 1 — Perplexity (Find sources):
'Find academic research papers and official reports about the impact of remittances

on consumer behavior, spending patterns, and economic development in Nepal.
Prioritize: Nepal Rastra Bank publications, World Bank reports, peer-reviewed papers
from 2015-2024. List each source with: authors, year, publication, key finding.'
STEP 2 — Claude (Create research outline):
'Using the research theme of remittance impact on Nepali consumer behavior, create
a detailed 3,000-word academic report outline with 6 major sections. Each section
should include: section title, 3-4 specific points to address, suggested sources
category (statistical data / case study / academic theory), and word allocation.'
STEP 3 — ChatGPT (Draft each section with guidance):
'Write Section 2: Literature Review for an academic report on remittance impact on
Nepal. Using the Permanent Income Hypothesis and consumption theories, explain how
academic literature predicts remittances affect household spending. 500 words.
Academic tone. APA in-text citations (I will add specific page numbers manually).'
STEP 4 — Claude (Review and improve full draft):
'Review this 3,000-word research report draft [PASTE]. Identify: logical gaps,

```
unsupported claims, weak transitions, redundant sections. Suggest
specific
improvements with revised text for the weakest 3 paragraphs.'
```

CHAPTER 11: IT Field Case Studies & Coding with AI

For BCA students, AI is not just a research tool — it is a pair programmer, a debugger, a documentation writer, and an architecture consultant. Master these workflows.

11.1 AI-Assisted Development Workflow

Professional developers who use AI coding tools report 30-55% productivity improvements, but the key insight is that AI makes better developers — not replacements. You still need to understand the code to use it effectively. AI-assisted coding follows this cycle:

Phase	AI Tool	What AI Helps With	What YOU Must Do
Planning	ChatGPT/Claude	Architecture suggestions, technology stack selection	Evaluate tradeoffs based on your requirements
Coding	GitHub Copilot/Cursor	Autocomplete functions, generate boilerplate code	Review every line; understand what it does
Debugging	ChatGPT/Claude	Explain error messages, suggest fixes	Reproduce the bug; verify the fix works
Testing	Claude/ChatGPT	Generate test cases, edge case identification	Run tests; understand what they're testing
Documentation	ChatGPT	Write docstrings, README files, API docs	Verify accuracy; add context AI doesn't know
Code Review	Claude	Identify code smells, security issues, optimizations	Implement and test suggestions carefully

11.2 Bad vs Good Coding Prompts — Python

Case: Building a Student Grade Calculator

❌ BAD CODING PROMPT	✅ PROFESSIONAL CODING PROMPT
Write Python code for grades. What you get: • Ambiguous — grades of what? • A basic if-else grade classifier • No input handling • No error handling • Not aligned to Nepal's grading system	Write Python 3.11 code for a KMC student grade management system. Requirements: (1) Input: student name, ID, 5 subject marks (0-100), (2) Calculate GPA using Nepal's Tribhuvan University grading scale (A=90-100, B+=80-89, B=70-79, C+=60-69, C=50-59,

• Unusable without major rewriting	D=40-49, F=0-39), (3) Output: grade report as table, (4) Save to CSV file with timestamp, (5) Handle invalid inputs gracefully. Add docstrings. Use pandas.

CLAUDE OUTPUT — Professional Grade System:

```
"""
KMC Student Grade Management System
Uses Tribhuvan University grading scale
Compatible with Python 3.11+
"""
import pandas as pd
from datetime import datetime
from pathlib import Path

# TU Grading Scale
GRADE_SCALE = [
    (90, 100, 'A',  4.0, 'Distinction'),
    (80,  89, 'B+', 3.7, 'Very Good'),
    (70,  79, 'B',  3.3, 'Good'),
    (60,  69, 'C+', 2.7, 'Satisfactory'),
    (50,  59, 'C',  2.3, 'Pass'),
    (40,  49, 'D',  1.7, 'Marginal Pass'),
    (0,   39, 'F',  0.0, 'Fail'),
]
```

```
def assign_grade(mark: float) -> tuple[str, float, str]:
    """Return (letter_grade, grade_point, description) for a mark."""
    for low, high, letter, gp, desc in GRADE_SCALE:
        if low <= mark <= high:
            return letter, gp, desc
    raise ValueError(f'Mark {mark} out of valid range 0-100')

def get_valid_mark(prompt: str) -> float:
    """Prompt until a valid 0-100 mark is entered."""
    while True:
        try:
            val = float(input(prompt))
            if 0 <= val <= 100: return val
            print('  ✗ Enter a number between 0 and 100.')
        except ValueError:
            print('  ✗ Invalid input — numbers only.')

# [Full implementation with CSV export, table display continues...]
```

11.3 Web Development — HTML/CSS Landing Page

Scenario: BCA Mini-Startup Landing Page

During the workshop, BCA students build a landing page for their startup using AI. Here's the professional approach:

PROFESSIONAL WEB DEV PROMPT:
Write clean, production-quality HTML5 and CSS3 for a single-page landing page for
'ParkSathi' — an AI parking app for Kathmandu. Requirements:
STRUCTURE: (1) Fixed navigation bar with logo and 3 nav links
(2) Hero section with tagline, sub-headline, and 2 CTA buttons
(3) Problem section with 3 statistic cards (Kathmandu parking stats)
(4) Features section with 3 feature cards with icons
(5) How It Works section with 3 numbered steps
(6) Footer with social links and copyright
STYLE: Modern, mobile-responsive, dark navy (#0A2342) and electric blue (#2E86AB),
clean sans-serif typography (Inter or system fonts), smooth hover animations,
card shadows, gradient hero background. NO external CSS frameworks.
CODE QUALITY: Semantic HTML5 elements, CSS custom properties for colors,
responsive using CSS Grid and Flexbox, media queries for mobile (max 768px),
comments explaining each major section, no JavaScript required for basic layout.

11.4 API Integration — Calling an AI API

For advanced BCA students, integrating an AI API into your own application is a valuable skill. This example shows how to call the OpenAI API from Python.

PROMPT FOR API INTEGRATION CODE:
`Write a complete Python script that uses the OpenAI API (gpt-4o-mini model) to build`
`a simple customer service chatbot for a Nepali e-commerce store.`
`Requirements:`
`1. Load API key from environment variable (never hardcode)`
`2. Maintain conversation history across multiple turns`
`3. System prompt: 'You are a helpful customer service agent for SastoBazar Nepal.`
`  Respond only in Nepali or English based on customer's language. Be concise.`
`  If asked about orders/tracking, ask for order number first.'`
`4. Handle API errors gracefully with user-friendly messages`
`5. Add rate limiting: max 10 messages per session`
`6. Log conversations to a text file with timestamps`
`7. Clean exit when user types 'exit' or 'quit'`
`Include: requirements.txt, setup instructions in comments, example .env file template`

11.5 Debugging with AI — The Detective Approach

Debugging is where AI assistance truly shines. Instead of spending hours searching Stack Overflow, you can paste your error and code directly into Claude or ChatGPT and get targeted help.

EFFECTIVE DEBUGGING PROMPT TEMPLATE:

I have a Python/JavaScript/[LANGUAGE] error I need help debugging.
LANGUAGE AND VERSION: Python 3.11 / Node.js 18 / etc.
WHAT I'M TRYING TO DO: [Describe the goal in 1-2 sentences]
WHAT HAPPENS INSTEAD: [Exact error message — copy/paste the full traceback]
MY CODE:
``
[Paste the relevant code — the function or section where the error occurs]
``
WHAT I'VE ALREADY TRIED:
[List any solutions you've attempted so the AI doesn't repeat them]
Please: (1) Explain what is causing this error in simple terms,
(2) Provide the corrected code with the fix highlighted,
(3) Explain WHY the fix works,
(4) Suggest how to avoid this type of error in the future

11.6 GitHub Copilot — AI in Your Code Editor

GitHub Copilot is an AI pair programmer that integrates directly into VS Code, JetBrains IDEs, and other editors. It reads your code and comments in real time and suggests completions. BCA students should set this up for all future projects.

Copilot Feature	How to Trigger	Best Practice
Line completion	Start typing — Copilot auto-suggests	Press Tab to accept, Esc to reject, Alt+] for next suggestion
Function generation	Write a descriptive function name and docstring	Be descriptive: def calculate_gpa_tribhuvan_scale(marks): → Copilot writes the body
Comment-driven code	Write # comment describing what you need	Write: # Create a function that reads student marks from CSV and returns GPA
Test generation	Open test file, type test_ → describe what to test	Be specific: test_invalid_mark_raises_value_error
Copilot Chat	Ctrl+I in VS Code for inline chat	Select code → 'Explain this', 'Refactor this', 'Add error handling'

11.7 Database Design with AI Assistance

DATABASE DESIGN PROMPT — Student Management System:
You are a senior database architect. Design a MySQL database schema for a
university student management system for Kathmandu Model College.
Requirements:
Entities: Students, Courses, Faculty, Enrollments, Grades, Departments
Include: (1) CREATE TABLE statements with all columns and data types,
(2) Primary keys and foreign keys with ON DELETE CASCADE where appropriate,
(3) Indexes on commonly queried columns (student_id, course_id, academic_year),
(4) At least 5 INSERT sample data rows per table,
(5) 5 useful SELECT queries (e.g., 'Get all students with GPA > 3.0 in BCA'),

(6) A GPA calculation view
Nepal context: Academic years run Jul-Jun; semester system; credit hours system;
TU grading scale; programs: BCA, BBA, BBM; batches by year of enrollment.
Use snake_case naming. Add SQL comments for each table.

CHAPTER 12: AI Tool Comparison Matrix

Not all AI tools are created equal. This chapter gives you a definitive reference for choosing the right tool for every task.

12.1 Master Comparison: All Tools

Capability	ChatGPT	Gemini	Claude	Perplexity	Gamma	Leonardo
Text Generation	★★★★★	★★★★☆	★★★★★	★★★★☆	★★★☆☆	✗
Research / Fact-finding	★★★☆☆	★★★★☆	★★★☆☆	★★★★★	✗	✗
Image Generation	★★★★☆	★★★☆☆	✗	✗	✗	★★★★★
Presentation Creation	✗	✗	✗	✗	★★★★★	✗
Code Generation	★★★★★	★★★★☆	★★★★★	★★★☆☆	✗	✗
Data Analysis	★★★★★	★★★☆☆	★★★★☆	✗	✗	✗
Long Document Processing	★★★☆☆	★★★★★	★★★★★	★★★☆☆	✗	✗
Real-time Web Info	★★★☆☆	★★★★★	★★☆☆☆	★★★★★	✗	✗
Free Tier Quality	★★★☆☆	★★★★☆	★★★★☆	★★★★☆	★★★☆☆	★★★★☆
Ease of Use	★★★★★	★★★★☆	★★★★★	★★★★★	★★★★★	★★★★☆

★★★★★ = Best in class | ★★★★☆ = Excellent | ★★★☆☆ = Good | ★★☆☆☆ = Limited | ✗ = Not applicable

12.2 Task-Based Tool Decision Guide

Your Task	Primary Tool	Supporting Tool	Why This Combination
Write a 2000-word report	Claude	Perplexity (for facts)	Claude for quality writing, Perplexity for verified data

Create a pitch deck	Gamma.app	Leonardo.ai (for logo)	Gamma builds structure, Leonardo creates visual brand
Research current events	Perplexity	Gemini (for depth)	Perplexity for facts + sources, Gemini for analysis
Build a web app	GitHub Copilot	Claude (for review)	Copilot in IDE, Claude for architecture and review
Analyze uploaded data	ChatGPT (Code Interp.)	Claude (for narrative)	ChatGPT runs analysis, Claude writes the story
Design a logo	Leonardo.ai	Canva (for text)	Leonardo generates the icon, Canva adds text/polish
Learn a new concept	ChatGPT or Claude	YouTube + Gemini	AI explains, then you verify with real resources
Write academic essay	Claude	Perplexity (sources)	Claude for writing quality, Perplexity for citations
Social media content	ChatGPT	Leonardo.ai (visuals)	ChatGPT for captions, Leonardo for images
Debug code	Claude or ChatGPT	GitHub Copilot	Paste error into Claude/GPT, Copilot for ongoing fixes

12.3 Pricing Comparison (as of 2025)

Tool	Free Tier	Paid Plan	What Paid Adds
ChatGPT	GPT-3.5 (limited)	Plus: $20/month (~NPR 2,700)	GPT-4o, DALL-E, Code Interpreter, Plugins
Google Gemini	Gemini 1.5 Flash free	Advanced: $19.99/month via Google One	Gemini Ultra, 1M context, priority speed
Claude	Claude 3.5 Sonnet free (limited)	Pro: $20/month (~NPR 2,700)	Higher limits, priority access, Projects feature
Perplexity	5 free Pro searches/day	Pro: $20/month (~NPR 2,700)	Unlimited searches, file uploads, Claude/GPT access
Gamma.app	400 AI credits (about 8 decks)	Plus: $8/month or Pro: $15/month	Unlimited decks, custom branding, analytics
Leonardo.ai	150 tokens/day (~30 images)	Apprentice: $10/month	750 tokens/day, priority queue, advanced models

Prices are indicative as of early 2025 and subject to change. Check official websites for current pricing.

12.4 Output Quality Comparison — Same Prompt, Different Tools

Test Prompt: 'Explain blockchain technology for a business student in Nepal. Give a local example.'

ChatGPT (GPT-4o) Response Quality:
Quality: ★★★★☆ | Style: Clear, structured, engaging
Approach: Explains blockchain using the analogy of a communal khata (ledger) in a
village cooperative, where every member has a copy and changes require consensus.
Local example: 'Imagine if Nepal's land registry (malpot) were on a blockchain —
no single government office could fraudulently alter ownership records, and any
citizen could verify land ownership in seconds.'
Verdict: Very good — Nepal-specific, accessible analogy, practical example

Claude Response Quality:
Quality: ★★★★★ | Style: Nuanced, comprehensive, well-reasoned
Approach: Starts with the core problem blockchain solves (trust without a central authority),
then builds up the concept layer by layer — distributed ledger → cryptography → consensus.
Nepal example: Discusses remittance corridors (Nepal receives ~26% GDP in remittances)
and how blockchain-based remittance (vs. traditional Western Union at 6-8% fee) could
save Nepali families billions annually. References actual companies attempting this.
Includes honest caveat about blockchain hype vs. genuine use cases.

Verdict: Best overall — deepest, most honest, most relevant to Nepal's economy

Gemini Response Quality:
Quality: ★★★★☆ | Style: Informative, well-cited
Approach: Clear technical explanation, cites specific Nepal government blockchain initiatives
(searches live web to find current data). Finds actual NRB mentions of digital currency research.
Local example: References Nepal Electricity Authority's pilot use of blockchain for
energy trading between households with solar panels.
Verdict: Best for current facts — finds real, recent Nepal-specific initiatives

CHAPTER 13: Advanced Techniques & The Future of AI

"The greatest danger in times of turbulence is not the turbulence — it is to act with yesterday's logic." — Peter Drucker

13.1 Advanced Prompting Techniques

Tree of Thought Prompting

An advanced version of Chain-of-Thought where you ask the AI to explore multiple reasoning paths simultaneously and then select the best one. Particularly effective for complex problems with multiple valid approaches.

TREE OF THOUGHT EXAMPLE:
I need to decide how to market my new trekking tour company in Nepal to international
tourists. Explore 3 different strategic approaches as separate 'branches':
Branch A: Digital-first strategy (social media, SEO, online booking)
Branch B: Partnership strategy (hotel tie-ups, embassy contacts, travel blogger outreach)
Branch C: Community strategy (local guides, village homestays, sustainable tourism cert)
For each branch, analyze: (1) Time to first booking, (2) Cost, (3) Scalability,
(4) Risks, (5) Best type of traveler attracted.
After exploring all three branches, synthesize: which elements would you combine

```
for a Year 1 strategy with NPR 5 lakh marketing budget?
```

Metacognitive Prompting

You ask the AI to reflect on its own reasoning and potential errors before presenting its answer. This significantly improves accuracy on complex analytical tasks.

METACOGNITIVE PROMPT:
Before you answer, take a moment to consider:
1. What assumptions am I making that might be wrong?
2. What information might I not have that would change my answer?
3. What is the strongest argument AGAINST my recommendation?
Now, with those considerations in mind, give me your recommendation for whether
a BCA student should learn Flutter or React Native for mobile app development in Nepal.

13.2 Building AI-Powered Workflows

The most powerful use of AI is not using a single tool, but building integrated workflows where multiple AI tools are chained together, each performing what it does best.

Workflow Name	Steps	Output
Research → Report	1. Perplexity (find data) → 2. Claude (analyze) → 3. ChatGPT (format) → 4. Gamma (present)	Professional research presentation
Idea → Brand	1. ChatGPT (name ideas) → 2. Perplexity (trademark check) → 3. Leonardo (logo) → 4. Canva (brand kit)	Complete startup brand identity
Data → Insight	1. Upload CSV to ChatGPT → 2. Code Interpreter (analysis) → 3. Claude (narrative) → 4. Gamma (deck)	Executive data presentation
Problem → Solution	1. Define problem to Claude → 2. Gemini (research solutions) → 3. ChatGPT (implementation plan)	Actionable solution roadmap

13.3 AI Agents: The Next Frontier

An AI agent is a system where an AI model can take sequences of actions autonomously to achieve a goal — browsing the web, writing and executing code, sending emails, and interacting with external services. While the tools you're learning today require a human at each step, agents reduce the need for human intervention.

Current Agent Examples

- AutoGPT / AgentGPT: Early agents that could autonomously research and execute multi-step tasks
- OpenAI Operator: Browses web and fills forms autonomously on your behalf
- Claude Computer Use: Controls a computer, clicks buttons, and completes tasks like a human assistant
- Custom GPTs with Actions: ChatGPT can call external APIs, search databases, and trigger workflows
- Cursor Composer: AI writes entire multi-file codebases, installs dependencies, and runs tests

13.4 The Future of AI in Nepal

Opportunities for Nepali Entrepreneurs

Nepal is in a unique position — AI tools are globally accessible, but local AI applications for Nepali problems are almost entirely undeveloped. This creates extraordinary opportunities for students who master AI now.

Opportunity Area	Problem to Solve	AI Tools Needed	Market Potential
Nepali NLP	Very few AI tools understand Nepali/Newari language	Custom LLM fine-tuning	Government, education, media sectors
AgriTech	Smallholder farmers lack access to expert advice	ML crop disease detection, chatbots	3M+ farming households
Legal Aid	Most Nepalis can't afford legal consultation	LLM trained on Nepal legal code	High urban demand
Healthcare Access	Doctor shortage in rural areas (1:2000 ratio)	Symptom checker + telemedicine AI	Massive government interest
Tourism AI	Poor digital experience for international trekkers	Multimodal AI guide apps	1M+ tourists/year
EdTech	Quality education not reaching all economic levels	Personalized learning AI in Nepali	Huge underserved market

13.5 Careers in AI — What to Learn Next

For BBA/BBM Students

- AI Product Management: Learn to build and manage AI-powered products — high demand, no coding required
- AI Marketing Specialist: Use AI tools for content, ads, analytics — the entire marketing stack is being transformed
- Data Analytics: SQL + Python basics + BI tools (Tableau, Power BI) + AI assistants = powerful career combination
- AI Business Strategy: Consultants who help traditional businesses adopt AI ethically and effectively

For BCA Students

- Machine Learning Engineer: Python + PyTorch/TensorFlow + mathematics + model deployment on cloud
- AI Application Developer: Build products using AI APIs (OpenAI, Anthropic, Google) — very high demand
- Data Engineer: Build the data pipelines that feed AI systems — foundational but critical skill
- Prompt Engineer / AI Systems Designer: Specialized role designing prompt architectures for enterprise AI
- MLOps Engineer: Deploy, monitor, and maintain ML models in production — bridging dev and operations

13.6 Building Your Personal AI Toolkit — Action Plan

Timeframe	Action	Goal
This Week	Complete all accounts: ChatGPT, Gemini, Claude, Perplexity, Gamma, Leonardo	Zero-friction access to all workshop tools
Week 2-4	Use AI for ONE real assignment — full RCTC prompt workflow	First hands-on application to coursework
Month 2	Master ONE tool deeply — use ChatGPT Code Interpreter OR Gamma for 5 projects	Develop expertise in at least one tool
Month 3	Build a simple AI-powered project (BCA: app with API; BBA: full market research)	Portfolio piece demonstrating AI competence
Month 4-6	Share learnings — teach a classmate, write a LinkedIn post, join AI communities	Teaching solidifies learning; visibility matters
Year 1	Start a small AI-enhanced side project or freelance service in Nepal	Apply learning to real economic opportunity

APPENDIX A: Prompt Templates & Cheat Sheets

Quick reference templates for every common student use case. Copy, customize, and use.

Business Writing Templates

TEMPLATE 1 — SWOT Analysis:
Act as a business analyst with expertise in [INDUSTRY] in Nepal.
Create a detailed SWOT analysis for [BUSINESS/IDEA].
Context: [2-3 sentences about the business and market]
Format: Table with 5 points per quadrant. Focus on 2025 Nepal market conditions.

TEMPLATE 2 — Business Plan Section:
You are a business plan writer. Write the [SECTION NAME] section of a business plan for
[BUSINESS NAME], a [business type] in [location], Nepal. Target customers: [describe].
Capital available: NPR [amount]. Tone: Professional, investor-facing. Length: [word count].

TEMPLATE 3 — Marketing Copy:
You are a copywriter specializing in [industry] marketing in Nepal.
Write [CONTENT TYPE: social post/email/ad copy] for [BRAND NAME] targeting [AUDIENCE].
Key message: [1 sentence]. Tone: [casual/professional/inspiring].
Include: [specific elements]. Length: [word/character limit].

Academic Templates

TEMPLATE 4 — Essay Introduction:
Write an academic introduction (150 words) for an essay titled '[TITLE]'.
Thesis statement: [YOUR THESIS]. Course: [COURSE NAME]. Level: [undergraduate/graduate].
Include: hook, background context, thesis preview. APA academic writing style.

TEMPLATE 5 — Literature Review:
Write a 400-word literature review section for a paper on '[TOPIC]'.
Cover: key theories, seminal studies, recent research (2018-2025), research gaps.
Use in-text citations as: (Author, Year) — I will fill in specific sources later.
Academic tone. Do NOT fabricate specific statistics — use general trends.

IT / Coding Templates

TEMPLATE 6 — Code Generation:
Write [LANGUAGE] [VERSION] code for: [TASK DESCRIPTION].
Input: [describe input format/source]. Output: [describe expected output].
Requirements: [list specific features]. Handle errors for: [list edge cases].
Add: docstrings, type hints, inline comments for complex sections.

TEMPLATE 7 — Code Explanation:
Explain the following [LANGUAGE] code to a [beginner/intermediate] programmer.

For each significant block: (1) What it does, (2) Why it's written this way,
(3) Any potential issues or improvements. Use simple analogies where helpful.
[PASTE CODE HERE]

APPENDIX B: Ethics, Hallucinations & Academic Integrity

This appendix consolidates all ethical guidelines for AI use at KMC. This is your reference when you are unsure whether a particular use of AI is acceptable.

The KMC AI Ethics Framework

Principle	What It Means in Practice
Transparency	When AI substantially helps with your work, disclose it. 'AI-assisted research' is honest and professional.
Verification	Never submit AI-generated facts without verifying them against primary sources.
Originality	Your analysis, arguments, and conclusions must be your own thinking — even if AI helped structure them.
Attribution	Cite sources found via AI the same way you'd cite sources found any other way.
Competence	Use AI to accelerate learning, not bypass it. You must understand what you submit.
Proportionality	More AI assistance = more your responsibility to verify and understand the output.

Hallucination Risk by Task Type

Task Type	Hallucination Risk	Verification Required
Explaining concepts and theories	Low — AI is well-trained on textbook knowledge	Light — cross-check unusual claims
Generating creative content (stories, slogans)	N/A — there's no 'wrong' answer	Check for accidental copyright issues
Specific statistics and data	VERY HIGH — AI invents numbers confidently	Always verify with primary/official sources
Research paper citations	EXTREME — AI fabricates papers convincingly	Always search Google Scholar to confirm

Current events (post training cutoff)	VERY HIGH — training data has a cutoff date	Use Perplexity or Gemini with web search
Code functionality	Medium — code may look correct but fail in practice	Always run and test every code snippet
Legal/medical/financial advice	High + Dangerous — may be convincingly wrong	Always consult qualified professionals

The 5-Question AI Ethics Check

Before submitting any AI-assisted work, ask yourself:

19. Did I understand what the AI produced, or am I just copying it?
20. Have I verified all factual claims against primary sources?
21. Does this work reflect MY thinking and analysis, or just AI's output?
22. Would I be comfortable explaining to my professor exactly how AI was used?
23. Have I added value beyond what AI alone produced?

If you answer YES to all 5, you are using AI ethically. If you answer NO to any, revise your approach before submitting.

Kathmandu Model College AI Workshop

Chaitra 2081 | KMC Department of BBA, BBM & BCA

"The goal is not to use AI instead of thinking — it's to use AI to think better."